PATENT COOPERATION TREATY

PCT

INTERNATIONAL SEARCH REPORT

(PCT Article 18 and Rules 43 and 44)

Applicant's or agent's file reference		FOR FURTHER ACTION
SONA-0013-WO		see Form PCT/ISA/220 as well as, where applicable, item 5 below.
International application No.	International filing date *(day/month/year)*	(Earliest) Priority Date *(day/month/year)*
PCT/US 22/46292	11 October 2022 (11.10.2022)	08 October 2021 (08.10.2021)
Applicant SONATUS, INC.		

This international search report has been prepared by this International Searching Authority and is transmitted to the applicant according to Article 18. A copy is being transmitted to the International Bureau.

This international search report consists of a total of 4 sheets.

☐ It is also accompanied by a copy of each prior art document cited in this report.

1. **Basis of the report**

   a. With regard to the **language**, the international search was carried out on the basis of:

   ☒ the international application in the language in which it was filed.

   ☐ a translation of the international application into _____ which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b)).

   b. ☐ This international search report has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43.6*bis*(a)).

   c. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, see Box No. I.

2. ☐ **Certain claims were found unsearchable** (see Box No. II).

3. ☒ **Unity of invention is lacking** (see Box No. III).

4. With regard to the **title**,

   ☒ the text is approved as submitted by the applicant.

   ☐ the text has been established by this Authority to read as follows:

5. With regard to the **abstract**,

   ☒ the text is approved as submitted by the applicant.

   ☐ the text has been established, according to Rule 38.2, by this Authority as it appears in Box No. IV. The applicant may, within one month from the date of mailing of this international search report, submit comments to this Authority.

6. With regard to the **drawings**,

   a. the figure of the **drawings** to be published with the abstract is Figure No. 7

   ☒ as suggested by the applicant.

   ☐ as selected by this Authority, because the applicant failed to suggest a figure.

   ☐ as selected by this Authority, because this figure better characterizes the invention.

   b. ☐ none of the figures is to be published with the abstract.

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International application No.
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Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:
because they relate to subject matter not required to be searched by this Authority, namely:

2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:

3. ☐ Claims Nos.:
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:
This application contains the following inventions or groups of inventions which are not so linked as to form a single general inventive concept under PCT Rule 13.1. In order for all inventions to be searched, the appropriate additional search fees must be paid.

Group I: Claims 1-20 & 39-60 are drawn to systems and methods for vehicle intrusion detection and automated intrusion response.

Group II: Claims 21-38 are drawn to a system for interpreting intrusion communications from a plurality of vehicles and providing an intrusion overview communication.

Group III: Claims 61-90 are drawn to a system for log monitoring and log analysis to detect a risk event in response to the log corpus.

--See extra sheet--

1. ☒ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.

2. ☐ As all searchable claims could be searched without effort justifying additional fees, this Authority did not invite payment of additional fees.

3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

4. ☐ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims; it is covered by claims Nos.:

Remark on Protest

☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.

☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.

☒ No protest accompanied the payment of additional search fees.

INTERNATIONAL SEARCH REPORT

International application No.
PCT/US 22/46292

A. CLASSIFICATION OF SUBJECT MATTER
IPC - INV. G07C 5/00, G07C 5/08 (2023.01)
ADD.
CPC - INV. G07C 5/008, G07C 5/0816, G07C 5/0808
ADD.
According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
See Search History document

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched
See Search History document

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
See Search History document

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X,D --- Y --- A	US 20210192867 A1 (Sonatus, Inc.) 24 June 2021 (24.06.2021) entire document especially Figs. 1-4, 5A, 5B, 6-210 & para [0011], [0223], [0225], [[0236], [0241], [0242], [0257], [0259], [0273], [0279], [0289, [0308], [0335], [0342], [0354], [0358]-[0360], [0362], [0381], [[0384], [0518], [0519], [0521],[0522], [0527], [0533], [0534], [0608], [0885], [0887]).	1-27, 29-60 --- 28 --- 61-90
Y	US 10,573,168 B1 (Johnson Controls Technology Company) 25 February 2020 (25.02.2020) entire document especially col. 22, lines 58-62 & col. 23, lines 31-33).	28
A	US 2019/0379683 A1 (NVIDIA Corporation) 12 December 2019 (12.12.2019) entire document especially Abstract & para [0040], [0048], [0198]).	61-90
A	US 2021/0197831 A1 (Hyundai Motor Company) 01 July 2021 (01.07.2021) entire document especially para [0073], [0091], [0166]).	61-90

☐ Further documents are listed in the continuation of Box C. ☐ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"D" document cited by the applicant in the international application
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
26 February 2023 (26.02.2023)	MAR 07 2023
Name and mailing address of the ISA/US Mail Stop PCT, Attn: ISA/US, Commissioner for Patents P.O. Box 1450, Alexandria, Virginia 22313-1450 Facsimile No. 571-273-8300	Authorized officer Kari Rodriquez Telephone No. PCT Helpdesk: 571-272-4300

Continuation of Box No. III, Observations where unity of invention is lacking:

The inventions listed as Groups I through III do not relate to a single general inventive concept under PCT Rule 13.1 because under PCT Rule 13.2 they lack the same or corresponding technical features for the following reasons:

Special Technical Features:

The special technical feature of Group I is a network intrusion detection component configured to detect an intrusion event in response to the network communications; and an intrusion response component configured to perform an automated intrusion response operation in response to the detected intrusion event, not required by Groups II & III.

The special technical feature of Group II is an intrusion processing component configured to identify an intrusion risk in response to the intrusion communications; and an intrusion reporting component configured to provide an intrusion overview communication in response to the intrusion risk, not required by Groups I & III.

The special technical feature of Group III is a log monitoring component configured to interpret a log corpus associated with at least one of the plurality of end points; a log analysis component configured to detect a risk event in response to the log corpus, not required by Groups I & II.

Common Technical Features:

Group I-III share the technical features of a system comprising a vehicle comprising a plurality of network zones, each network zone comprising a plurality of end points; and a controller comprising a monitoring component configured to interpret communications, an analysis component configured to detect an event, and a response component configured to provide a response operation in response to the detected event. However, the shared technical features do not represent a contribution over

US 20210192867 A1 (Sonatus, Inc.) (hereinafter Sonatus) 24 June 2021 (24.06.2021)

Regarding claim 1, Sonatus discloses a system (system; para [0553]), comprising: a vehicle comprising a plurality of network zones (a vehicle with one or more network zones; para [0553]), each network zone comprising a plurality of end points (providing end points on at least one network zone of a vehicle; configuration of network zones and/or end points on the vehicle; end points of at least one network zone, e.g., first network zone and second network zone; para [0019], [0519], [0521]); a controller (controllers; para [0553]), comprising: a network monitoring component configured to interpret network communications associated with at least one of the network zones (intrusion detection systems (IDS) across networks of a mobile application; collect vehicle data from one or more end points of at least one network zone, e.g., first network zone and second network zone in the vehicle in response to the parsed policy data; Figs. 1-4, 5A, 5B, 6-210 & para [0273]); a network intrusion detection component configured to detect an intrusion event in response to the network communications (intrusion detection systems (IDS) across networks of a mobile application; centralized intrusion detection system, where signals/packets and messages are analyzed and processed at the same location; para [0273], [0362]); and an intrusion response component configured to perform an intrusion response operation in response to the detected intrusion event (detecting an incident, and notifying a user about the incident; notification of the incident may further include additional information, such as pictures, video clips, audio information, incident determination descriptions, e.g., sensor/actuator value indicating intrusion; para [0885]).

Thus, unity of invention is lacking under PCT Rule 13.1 because Groups I-III do not share a same or corresponding special technical feature that would provide a unifying contribution over the prior art.

PATENT COOPERATION TREATY

From the INTERNATIONAL SEARCHING AUTHORITY

To: J. STEPHEN WILLS
GTC LAW GROUP PC & AFFILIATES
ONE UNIVERSITY AVE., STE. 302B
WESTWOOD, MA 02090

PCT

NOTIFICATION OF TRANSMITTAL OF THE INTERNATIONAL SEARCH REPORT AND THE WRITTEN OPINION OF THE INTERNATIONAL SEARCHING AUTHORITY, OR THE DECLARATION

(PCT Rule 44.1)

	Date of mailing *(day/month/year)* MAR 07 2023
Applicant's or agent's file reference SONA-0013-WO	**FOR FURTHER ACTION** See paragraphs 1 and 4 below
International application No. PCT/US 22/46292	International filing date *(day/month/year)* 11 October 2022 (11.10.2022)
Applicant SONATUS, INC.	

1. ☒ The applicant is hereby notified that the international search report and the written opinion of the International Searching Authority have been established and are transmitted herewith.

   Filing of amendments and statement under Article 19:
   The applicant is entitled, if he so wishes, to amend the claims of the international application (see Rule 46):

   When? The time limit for filing such amendments is normally two months from the date of transmittal of the international search report.

   How? Directly to the International Bureau preferably through ePCT, or on paper to:
   The International Bureau of WIPO, 34, chemin des Colombettes, 1211 Geneva 20, Switzerland

   For more detailed instructions, see the *PCT Applicant's Guide*, International Phase, paragraphs 9.004 – 9.011.

2. ☐ The applicant is hereby notified that no international search report will be established and that the declaration under Article 17(2)(a) to that effect and the written opinion of the International Searching Authority are transmitted herewith.

3. ☐ **With regard to any protest** against payment of (an) additional fee(s) under Rule 40.2, the applicant is notified that:

   ☐ the protest together with the decision thereon has been transmitted to the International Bureau together with any request to forward the texts of both the protest and the decision thereon to the designated Offices.

   ☐ no decision has been made yet on the protest; the applicant will be notified as soon as a decision is made.

4. **Reminders**

   The applicant may **submit comments on an informal basis on the written opinion of the International Searching Authority** to the International Bureau. These comments will be made available to the public after international publication. The International Bureau will send a copy of such comments to all designated Offices unless an international preliminary examination report has been or is to be established.

   Shortly after the expiration of **18 months from the priority date, the international application will be published** by the International Bureau. If the applicant wishes to avoid or postpone publication, a notice of withdrawal of the international application, or of the priority claim, must reach the International Bureau before the completion of the technical preparations for international publication (Rules 90*bis*.1 and 90*bis*.3).

   Within **19 months** from the priority date, but only in respect of some designated Offices, a demand for international preliminary examination must be filed if the applicant wishes to postpone the entry into the national phase **until 30 months** from the priority date (in some Offices even later); otherwise, the applicant must, **within 20 months** from the priority date, perform the prescribed acts for **entry into the national phase** before those designated Offices. In respect of other designated Offices, the time limit of **30 months** (or later) will apply even if no demand is filed within 19 months. For details about the applicable time limits, Office by Office, see www.wipo.int/pct/en/texts/time_limits.html and the *PCT Applicant's Guide*, National Chapters.

   Within **22 months from the priority date, the applicant may request that a supplementary international search be carried out** by a different International Searching Authority that offers this service (Rule 45*bis*.1). The procedure for requesting supplementary international search is described in the *PCT Applicant's Guide*, International Phase, paragraphs 8.006-8.032.

Name and mailing address of the ISA/US Mail Stop PCT, Attn: ISA/US Commissioner for Patents P.O. Box 1450, Alexandria, Virginia 22313-1450 Facsimile No. 571-273-8300	Authorized officer Kari Rodriquez Telephone No. PCT Helpdesk: 571-272-4300

Form PCT/ISA/220 (revised January 2020)